

Data Structures Homework - 1

1. Sum of Two Matrices

```
#include <stdio.h>

int main() {
    int r, c, i, j;
    printf("Enter rows and columns: ");
    scanf("%d%d", &r, &c);

    int a[r][c], b[r][c], sum[r][c];

    printf("Enter first matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&a[i][j]);

    printf("Enter second matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&b[i][j]);

    printf("Sum Matrix:\n");
    for(i=0;i<r;i++) {
        for(j=0;j<c;j++) {
            sum[i][j]=a[i][j]+b[i][j];
            printf("%d ",sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output :

```
Enter rows and columns: 3
3
Enter first matrix:
12
23
```

```
34
45
56
67
8
8
7
Enter second matrix:
23
32
43
4
65
76
87
98
6
Sum Matrix:
35 55 77
49 121 143
95 106 13
```

=== Code Execution Successful ===

2. Reverse an Array

```
#include <stdio.h>

int main() {
    int n,i;
    printf("Enter size: ");
    scanf("%d",&n);

    int a[n];

    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
```

```
    printf("Reverse Order:\n");
    for(i=n-1;i>=0;i--)
        printf("%d ",a[i]);

    return 0;
}
```

Output :

```
Enter size: 5
Enter elements:
23
45
56
7
8
Reverse Order:
8 7 56 45 23
```

=== Code Execution Successful ===

3. Count Duplicate Elements

```
#include <stdio.h>

int main() {
    int n,i,j,count=0;
    printf("\n Enter the N value :");
    scanf("%d",&n);

    int a[n],check[n];
    printf("Enter the array Elements : \n");
    for(i=0;i<n;i++) {
        scanf("%d",&a[i]);
        check[i]=0;
    }

    for(i=0;i<n;i++) {
```

```

        if(check[i]) continue;
        for(j=i+1;j<n;j++) {
            if(a[i]==a[j]) {
                check[j]=1;
                count++;
            }
        }
    }

    printf("Duplicate Elements = %d",count);
    return 0;
}

```

Output :

```

Enter the N value :5
Enter the array Elements :
23
23
45
45
45
Duplicate Elements = 3

```

=== Code Execution Successful ===

4. Print Unique Elements

```

#include <stdio.h>

int main() {
    int n,i,j,flag;
    printf("\n Enter the N value :");
    scanf("%d",&n);
    int a[n];
    printf("Enter the array Elements : \n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
}

```

```

printf("Unique Elements:\n");

for(i=0;i<n;i++) {
    flag=1;
    for(j=0;j<n;j++) {
        if(i!=j && a[i]==a[j]) {
            flag=0;
            break;
        }
    }
    if(flag)
        printf("%d ",a[i]);
}
return 0;
}

```

Output :

```

Enter the N value :5
Enter the array Elements :
34
45
45
56
5
Unique Elements:
34 56 5

```

=== Code Execution Successful ===

5. Insert into Sorted Array

```

#include <stdio.h>

int main() {
    int n,i,pos,value;
    printf("\n Enter the N value :");
    scanf("%d",&n);
}

```

```

int a[100];
printf("Enter the array Elements : \n");
for(i=0;i<n;i++)
    scanf("%d",&a[i]);

printf("Enter value: ");
scanf("%d",&value);

for(i=0;i<n;i++) {
    if(a[i]>value)
        break;
}
pos=i;

for(i=n;i>pos;i--)
    a[i]=a[i-1];

a[pos]=value;
n++;

for(i=0;i<n;i++)
    printf("%d ",a[i]);

return 0;
}

```

Output :

```

Enter the N value :5
Enter the array Elements :
34
5
6
7
78
Enter value: 45
34 5 6 7 45 78

```

=== Code Execution Successful ===

6. Insert into Unsorted Array

```
#include <stdio.h>

int main() {
    int n,i,pos,value;
    printf("\n Enter the N value :");
    scanf("%d",&n);

    int a[100];
    printf("Enter the array Elements : \n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Enter position: ");
    scanf("%d",&pos);

    printf("Enter value: ");
    scanf("%d",&value);

    for(i=n;i>=pos;i--)
        a[i]=a[i-1];

    a[pos-1]=value;
    n++;

    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    return 0;
}
```

Output :

```
Enter the N value :5
Enter the array Elements :
34
45
56
67
78
```

Enter position: 4
Enter value: 65
34 45 56 65 67 78

=== Code Execution Successful ===

7. Matrix Multiplication

```
#include <stdio.h>

int main()
{
    int r1, c1, r2, c2, i, j, k;

    printf("Enter rows and columns of first matrix: ");
    scanf("%d%d", &r1, &c1);

    printf("Enter rows and columns of second matrix: ");
    scanf("%d%d", &r2, &c2);

    if (c1 != r2)
    {
        printf("Matrix multiplication is not possible.");
        return 0;
    }

    int a[r1][c1], b[r2][c2], c[r1][c2];

    printf("Enter first matrix elements:\n");
    for (i = 0; i < r1; i++)
    {
        for (j = 0; j < c1; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Enter second matrix elements:\n");
    for (i = 0; i < r2; i++)
    {
```



```

        for (j = 0; j < c2; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for (i = 0; i < r1; i++)
    {
        for (j = 0; j < c2; j++)
        {
            c[i][j] = 0;
            for (k = 0; k < c1; k++)
            {
                c[i][j] += a[i][k] * b[k][j];
            }
        }
    }

    printf("Product Matrix:\n");

    for (i = 0; i < r1; i++)
    {
        for (j = 0; j < c2; j++)
        {
            printf("%d ", c[i][j]);
        }
        printf("\n");
    }

    return 0;
}

```

Output :

Enter rows and columns of first matrix: 3

3

Enter rows and columns of second matrix: 3

3

Enter first matrix elements:

1

2

```
3
4
5
6
7
8
9
Enter second matrix elements:
9
8
7
6
5
4
3
2
1
Product Matrix:
30 24 18
84 69 54
138 114 90
```

=== Code Execution Successful ===

8. Matrix Transpose

```
#include <stdio.h>

int main()
{
    int r, c, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d%d", &r, &c);

    int a[r][c];

    printf("Enter matrix elements:\n");
```

```

for (i = 0; i < r; i++)
{
    for (j = 0; j < c; j++)
    {
        scanf("%d", &a[i][j]);
    }
}

printf("Transpose Matrix:\n");

for (i = 0; i < c; i++)
{
    for (j = 0; j < r; j++)
    {
        printf("%d ", a[j][i]);
    }
    printf("\n");
}

return 0;
}

```

Output :

```

Enter number of rows and columns: 3
3
Enter matrix elements:
11
22
33
44
55
6
77
88
99
Transpose Matrix:
11 44 77
22 55 88
33 6 99

```

=== Code Execution Successful ===

9. Right Diagonal Sum

```
#include <stdio.h>

int main()
{
    int n, i, j, sum = 0;

    printf("Enter the order of matrix: ");
    scanf("%d", &n);

    int a[n][n];

    printf("Enter matrix elements:\n");

    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    for (i = 0; i < n; i++)
    {
        sum += a[i][n - 1 - i];
    }

    printf("Right diagonal sum = %d", sum);

    return 0;
}
```

Output :

Enter the order of matrix: 3
Enter matrix elements:

9
8
7
6
5
4
3
2
1

Right diagonal sum = 15

=== Code Execution Successful ===

10. Check Identity Matrix

```
#include <stdio.h>

int main()
{
    int n, i, j, flag = 1;

    printf("Enter the order of matrix: ");
    scanf("%d", &n);

    int a[n][n];

    printf("Enter matrix elements:\n");

    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
```

```

    {
        if (i == j && a[i][j] != 1)
        {
            flag = 0;
        }

        if (i != j && a[i][j] != 0)
        {
            flag = 0;
        }
    }
}

if (flag == 1)
    printf("It is an Identity Matrix.");
else
    printf("It is not an Identity Matrix.");

return 0;
}

```

Output :

Enter the order of matrix: 3

Enter matrix elements:

1

2

3

2

1

4

3

4

1

It is not an Identity Matrix.

=== Code Execution Successful ===

11. Count Vowels, Consonants, Digits and White Spaces

```

#include <stdio.h>

int main()
{
    char str[100];
    int i, vowels = 0, consonants = 0, digits = 0, spaces = 0;

    printf("Enter a string: ");
    scanf("%[^\n]", str);

    for (i = 0; str[i] != '\0'; i++)
    {
        if ((str[i] >= 'A' && str[i] <= 'Z') || (str[i] >= 'a' && str[i] <= 'z'))
        {
            char ch = str[i];

            if (ch >= 'A' && ch <= 'Z')
                ch = ch + 32;

            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
                vowels++;
            else
                consonants++;
        }
        else if (str[i] >= '0' && str[i] <= '9')
        {
            digits++;
        }
        else if (str[i] == ' ')
        {
            spaces++;
        }
    }

    printf("Vowels = %d\n", vowels);
    printf("Consonants = %d\n", consonants);
    printf("Digits = %d\n", digits);
    printf("White Spaces = %d\n", spaces);

    return 0;
}

```

Output :

Enter a string: sripriyan
Vowels = 3
Consonants = 6
Digits = 0
White Spaces = 0

=== Code Execution Successful ===

12. Frequency of a Character

```
#include <stdio.h>

int main()
{
    char str[100], ch;
    int i, count = 0;

    printf("Enter a string: ");
    scanf("%s", str);

    printf("Enter the character: ");
    scanf(" %c", &ch);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ch)
        {
            count++;
        }
    }

    printf("Frequency of %c = %d", ch, count);

    return 0;
}
```

Output :

Enter a string: onepiece

Enter the character: e

Frequency of e = 3

=== Code Execution Successful ===

13. Count Words in a String

```
#include <stdio.h>
```

```
int main()
{
    char str[100];
    int i, words = 1;

    printf("Enter a string: ");
    scanf("%[^\\n]", str);

    for(i = 0; str[i] != '\\0'; i++)
    {
        if(str[i] == ' ')
        {
            words++;
        }
    }

    printf("Number of words = %d", words);

    return 0;
}
```

Output :

Enter a string: hi im sripriyan

Number of words = 3

=== Code Execution Successful ===

14. Sort String Array

```
#include <stdio.h>
#include <string.h>

int main()
{
    int n, i, j;
    char str[20][50], temp[50];

    printf("Enter the number of strings: ");
    scanf("%d", &n);

    printf("Enter the strings:\n");

    for(i = 0; i < n; i++)
    {
        scanf(" %[^\n]", str[i]);
    }

    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(strcmp(str[i], str[j]) > 0)
            {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }

    printf("Strings in Sorted Order:\n");

    for(i = 0; i < n; i++)
    {
        printf("%s\n", str[i]);
    }

    return 0;
}
```

Output :

Enter the number of strings: 5

Enter the strings:

mani

akil

bala

sri

thiru

Strings in Sorted Order:

akil

bala

mani

sri

thiru

=== Code Execution Successful ===

15. Toggle Case of a Sentence

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char str[100];
```

```
    int i;
```

```
    printf("Enter a sentence: ");
```

```
    scanf("%[^\n]", str);
```

```
    for(i = 0; str[i] != '\0'; i++)
```

```
    {
```

```
        if(str[i] >= 'A' && str[i] <= 'Z')
```

```
        {
```

```
            str[i] = str[i] + 32;
```

```
        }
```

```
        else if(str[i] >= 'a' && str[i] <= 'z')
```

```
        {
```

```
            str[i] = str[i] - 32;
```

```
    }  
}  
  
printf("Sentence after toggling:\n%s", str);  
  
return 0;  
}
```

Output :

```
Enter a sentence: SrIpRiYaN  
Sentence after toggling:  
sRiPrIyAn
```

=== Code Execution Successful ===